

CHAPTER 4

EPIC 1: MODEL SELECTION AND ARCHITECTURE

4.1 Introduction

Model selection is a crucial phase in the EduGenie project. Appropriate AI models improve the accuracy, quality, and reliability of educational responses. EduGenie integrates advanced language models to provide explanations, question answering, quiz generation, summarization, and learning recommendations.

4.2 Need for AI Models

Traditional educational applications depend on static content. Generative AI enables intelligent responses, contextual understanding, and personalized learning. The selected models support natural language processing and content generation for students.

4.3 Google Gemini

Google Gemini is used as the primary AI model for answering questions, generating quizzes, summarizing content, and recommending learning paths. It provides high-quality responses with contextual understanding and supports a wide range of academic topics.

4.4 LaMini-Flan-T5

LaMini-Flan-T5 is a lightweight transformer model suitable for generating simple explanations. It is efficient, fast, and useful for explaining concepts in an easy-to-understand manner.

4.5 Model Comparison

Google Gemini offers superior reasoning, content generation, and conversational capabilities, while LaMini-Flan-T5 provides lightweight inference and simplified explanations. Combining both models improves overall application performance.

4.6 System Architecture

The architecture consists of an HTML/CSS frontend, FastAPI backend, AI processing layer, and response generation layer. User requests are routed through FastAPI to the appropriate AI model, and generated responses are returned to the interface.

4.7 Advantages

High response accuracy, modular implementation, scalable architecture, fast processing, easy maintenance, and seamless AI integration are major advantages of the selected architecture.

4.8 Chapter Summary

This chapter discussed the AI models selected for EduGenie, their roles, system architecture, and the benefits of integrating Google Gemini and LaMini-Flan-T5 within the FastAPI framework.